>hell

, world

a getting-started zine
for a box that plays text.

x hit . rest 0-9 how much _ hold [xx] split <a b> alternate x%30 maybe /2 *2 speed !4 end

>what

? what? try: help

a keyboard, a screen that
needs daylight, a clock.

you type lines. a line is
a lane. lanes play: drums,
synths, filters, pictures.

no mouse. no menu. no mode.
the page you are editing
is the page that plays.

```
laptop sets taught crowds
to read projected code as
proof that it was live
[21][4][3].
here the gesture and the
sound are one thing to
watch [19][15][2].
```

an esp32-s3 and a 400x300 reflective panel, one bit, no backlight. 96 ticks a beat, 16 lanes, 34 verbs. usb midi jitter at the host: 0.03 ms, measured. it makes no sound itself: it sends midi, osc and pictures to what does.

>hello

? hello? try: help

```
>kick 9...8...9...8...
```

velocity is ink

```
>play
```

ctrl+enter runs the line
enter makes a line
again, while it plays:
quiet. again: back.

it sends midi, not sound.
>usb on to a computer:
 the deck restarts
 as a midi device.
>din 17 to a synth, with
 no computer. build
 the cable first.

drums on channel 10, bass 1, lead 2, pad 3, arp 4: the names are lines in the boot document. change them there.

x is a hit. . is a rest.
now form a band [13].

>grammar

? grammar? try: help

x	a hit
.	a rest
0-9	how much
_	hold the last one
[xx]	two in one step
[0,4,7]	all at once
<a b>	one each time round
x%30	30 times in 100
/2 *2	this's lane's speed
!4	four times, then end

a line is as long as it
is. 16 against 12 comes
home every 3 bars. that
is all polyrhythm is.

struct writes bd ~ sd ~ where this writes x.x. and 51 of 51 corpus patterns play what struct 1.2.6 plays, note for note [18][12]. what the deck will not play, it refuses, and boxes the character:

```
>hat x...□...  X is gone: 9 is loud
>hat x...?...  ? is gone: x%50 is maybe
>hat x-□-      - is not a rest here: .
```


>names

? names? try: help

the names are yours:

```
>conga = note 63
>strings = voice 3 ch 5
>fx = cc 20 ch 2
>conga =          gone.

anything drives anything:
>route cut kick
>route disc kick
>tom x.x.x.x. !2
>route crash tom:end
```

a lane that ends can start
another. no timeline: an
arrangement is endings.

>knob1 = knob and >oso in 9000: a phone, a laptop or another deck sends /deck/knob1 and >route out knob1 follows it. the value lives here, so losing the phone loses nothing.

the deck talks back:

- _ needs a note before it
- a chord goes in [1: [0,4,7]
- a name is letters: conga
- 16 lanes is all there is.
- run cannot run itself

>help

? 34 commands - the one line here it knows

[1] aaron s, blackwell af 2013. from sonic pi to overtone. farm'13.
[2] berthaut f et al 2013. rouages: revealing the mechanisms of digital musical instruments to the audience. nime'13.
[3] burland k, mclean a 2016. understanding live coding events. ijpadm 12(2).
[4] collins n, mclean a, rohrhuber j, ward a 2003. live coding in laptop performance. organised sound 8(3).
[5] collins n, mclean a 2014. algorave. nime'14.
[6] eno b 1996. generative music 1. sseyo koan.
[7] gibson w 1984. neuromancer. ace.
[8] jack o 2017. hydra. software; iclc'19 demo.
[9] lovelace aa 1843. notes to menabrea. scientific memoirs 3.
[10] mathews mv 1963. the digital computer as a musical instrument. science 142.
[11] mccartney j 2002. rethinking the computer music language: supercollider. cmj 26(4).
[12] mclean a 2014. making programming languages to dance to: live coding with tidal. farm'14.
[13] moon t 1977. sideburns 1, p2.
[14] obarski k 1987. ultimate soundtracker; goriunova o 2012, p102.
[15] reeves s, benford s, o'malley c, fraser m 2005. designing the spectator experience. chi'05.
[16] reich s 1968. music as a gradual process.
[17] riley t 1964. in c. sf tape music center.
[18] roos f, mclean a 2023. strudel: live coding patterns on the web. iclc'23.
[19] schloss wa 2003. using contemporary technology in live performance: the dilemma of the performer. jnmr 32(3).
[20] spiegel l 1986. music mouse.
[21] toplap 2004. manifestodraft; ward a et al, read_me.
[22] toussaint g 2005. the euclidean algorithm generates traditional musical rhythms. bridges.
[23] weingart w 2000. typography. lars muller.
[24] xenakis i 1971. formalized music. indiana up.

set in the deck's own 12x24 and 6x12 faces; one bit; every picture drawn by its engine. a deck after gibson [7], a grid after weingart, sometimes called swiss punk [23]. full references: zine/README.md. copy it, fold it, pass it on.

>room

? room? try: help

1843	lovelace	an engine could compose [9]
1963	mathews	the computer, an instrument [10]
1964	riley	in c: 53 phrases, repeated [17]
1968	reich	music as a gradual process [16]
1971	xenakis	formalized music [24]
1986	spiegel	music mouse [20]
1987	obarski	the tracker [14]
1996	eno	generative music 1 [6]
2002	mccartney	supercollider 3 [11]
2003	collins et al	live coding [4]
2004	toplap	show us your screens [21]
2005	toussaint	euclid: open here still [22]
2013	aaron, blackwell	sonic pi [1]
2014	mclean	tidal [12]
2014	collins, mclean	algorave [5]
2017	jack	hydra [8]
2023	roos, mclean	strudel [18]
2026	you	this box

for anyone who lost a
night to a tracker, a
patch cable or a
semicolon.

on the deck: ground, lift,
orbitals. open one, run it
top to bottom. every set
ends in silence.

>pictures

? pictures? try: help

six fields, ten operators,
no shapes.

fields disc box turn ramp grid noise
levels mask edge
bends echo move spin warp grow thin flip fold

orbitals, drawn by the deck: a planet, x at 16 steps and y at 12, round a sun on the beat; night, day, eclipse. echo keeps the trail: feedback is the trick [8].